

Cuaderno de Prácticas

MATEMÁTICAS DISCRETAS

Grado en Ingeniería Informática

CURSO 2025/26

Datos personales

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Grupo de teoría : A

Grupo de prácticas : 4

Práctica I

Ejercicio I . 6

Texto del ejercicio 1

Solución :

Ejercicio 2.1

Calcular con 5 y 10 cifras significativas :

a) $3 (1 + 4) - 2^2 \times 5 - 5^{1/5}$

In[]:=* `N[3(1+4)-22×5-51/5,10]`

Out[]=*

-6.379729661

b) $\frac{1}{2} + \frac{1}{3} + \frac{1}{5}$

```
In[*]:= N[ $\frac{1}{2} + \frac{1}{3} + \frac{1}{5}$ , 10]
```

```
Out[*]= 1.033333333
```

c) $\frac{\sqrt{5}}{\sqrt[3]{3}}$

```
In[*]:= N[ $\frac{\sqrt{5}}{\sqrt[3]{3}}$ , 10]
```

```
Out[*]= 1.550402942
```

i) 2^{1000}

```
In[*]:= N[ $2^{1000}$ , 10]
```

```
Out[*]= 1.071508607  $\times 10^{301}$ 
```

Ejercicio 2.3

Sea x el número de tu DNI e y el año en que naciste.

```
In[*]:= x=26268082;  
y=2004;
```

a) Comprobar si x es primo

```
In[*]:= PrimeQ[x]
```

```
Out[*]= False
```

b) Calcular el cociente y el resto de dividir x entre y

```
In[ ]:= "Cociente"
        Quotient[x,y]
        "Resto"
        Mod[x,y]
```

```
Out[ ]=
Cociente
```

```
Out[ ]=
13 107
```

```
Out[ ]=
Resto
```

```
Out[ ]=
1654
```

```
In[ ]:= Mod[x,y]
```

```
Out[ ]=
1654
```

c) Calcular una aproximación decimal con 20 cifras decimales de la raíz cuadrada de x

```
In[ ]:= N[Sqrt[x],20]
```

```
Out[ ]=
5125.2397017115209262
```

d) Calcular el entero más próximo al número $(\pi y - e)/x$

```
In[ ]:= Round[ $\frac{\pi * y - e}{x}$ ]
```

```
Out[ ]=
0
```

e) Calcular el número de Fibonacci del día del mes en que naciste

```
In[ ]:= Fibonacci[6]
```

```
Out[ ]=
8
```

Práctica 2

Ejercicio 3.2

Crear una tabla como en el ejercicio 3.4 . cuya primera fila esté formada por los cinco primeros múltiplos positivos del día del mes en que naciste, la segunda fila por sus cubos y la tercera por la potencia quinta de dichos números

```
In[ ]:= tabla=Table[ (3*j)^i,{i,1,3},{j,1,5}];  
TableForm[tabla]
```

```
Out[ ]//TableForm=
```

3	6	9	12	15
9	36	81	144	225
27	216	729	1728	3375

Ejercicio 3.4

a) Crear un vector cuyos coeficientes sean los números menores que 100 y múltiplos de 13

```
In[ ]:= Table[i,{i, 13, 100,13}]
```

```
Out[ ]:=
```

{13, 26, 39, 52, 65, 78, 91}

b) Crear un vector que contenga los cuadrados de los múltiplos de 13 menores que tu año de nacimiento al cuadrado

```
In[ ]:= n=2004^2  
Table[ (13*i)^2,{i,153}]
```

```
Out[ ]:=
```

4 016 016

```
Out[ ]:=
```

{169, 676, 1521, 2704, 4225, 6084, 8281, 10816, 13689, 16900, 20449, 24336, 28561, 33124, 38025, 43264, 48841, 54756, 61009, 67600, 74529, 81796, 89401, 97344, 105625, 114244, 123201, 132496, 142129, 152100, 162409, 173056, 184041, 195364, 207025, 219024, 231361, 244036, 257049, 270400, 284089, 298116, 312481, 327184, 342225, 357604, 373321, 389376, 405769, 422500, 439569, 456976, 474721, 492804, 511225, 529984, 549081, 568516, 588289, 608400, 628849, 649636, 670761, 692224, 714025, 736164, 758641, 781456, 804609, 828100, 851929, 876096, 900601, 925444, 950625, 976144, 1002001, 1028196, 1054729, 1081600, 1108809, 1136356, 1164241, 1192464, 1221025, 1249924, 1279161, 1308736, 1338649, 1368900, 1399489, 1430416, 1461681, 1493284, 1525225, 1557504, 1590121, 1623076, 1656369, 1690000, 1723969, 1758276, 1792921, 1827904, 1863225, 1898884, 1934881, 1971216, 2007889, 2044900, 2082249, 2119936, 2157961, 2196324, 2235025, 2274064, 2313441, 2353156, 2393209, 2433600, 2474329, 2515396, 2556801, 2598544, 2640625, 2683044, 2725801, 2768896, 2812329, 2856100, 2900209, 2944656, 2989441, 3034564, 3080025, 3125824, 3171961, 3218436, 3265249, 3312400, 3359889, 3407716, 3455881, 3504384, 3553225, 3602404, 3651921, 3701776, 3751969, 3802500, 3853369, 3904576, 3956121}

c) Crear una tabla de dos filas cuya primera fila sean los números impares comprendidos entre el 20 y 36 y la segunda el resultado de sumar a la fila anterior tu número de nacimiento (ver ejercicio 3.3.)

```
In[*]:= dia=3;
t=Table[19+(2*j)+dia*(i-1),{i,2},{j,9}];
TableForm[t]
```

```
Out[*]//TableForm=
 21   23   25   27   29   31   33   35   37
 24   26   28   30   32   34   36   38   40
```

d) Crear una matriz 10 x 10 que contenga un 1 en las componentes a tales que $i+j$ es par y -1 en aquellas en las que $i+j$ es impar

```
In[*]:= t=Table[If[Mod[i+j,2]==0,1,-1],{i,10},{j,10}];
TableForm[t]
```

```
Out[*]//TableForm=
 1   -1   1   -1   1   -1   1   -1   1   -1
-1   1   -1   1   -1   1   -1   1   -1   1
 1   -1   1   -1   1   -1   1   -1   1   -1
-1   1   -1   1   -1   1   -1   1   -1   1
 1   -1   1   -1   1   -1   1   -1   1   -1
-1   1   -1   1   -1   1   -1   1   -1   1
 1   -1   1   -1   1   -1   1   -1   1   -1
-1   1   -1   1   -1   1   -1   1   -1   1
 1   -1   1   -1   1   -1   1   -1   1   -1
-1   1   -1   1   -1   1   -1   1   -1   1
```

e) Crear una tabla que simbolice un tablero de ajedrez de manera que en las casillas negras aparezca un 0 y en las casillas blancas aparezcan un 1

```
In[*]:= t=Table[If[Mod[i+j,2]==0,0,1],{i,8},{j,8}];
TableForm[t]
```

```
Out[*]//TableForm=
 0   1   0   1   0   1   0   1
 1   0   1   0   1   0   1   0
 0   1   0   1   0   1   0   1
 1   0   1   0   1   0   1   0
 0   1   0   1   0   1   0   1
 1   0   1   0   1   0   1   0
 0   1   0   1   0   1   0   1
 1   0   1   0   1   0   1   0
```

Práctica 3

De archivo de practicas el capitulo 4. Cada ejercicio hay que hacerlo con Do, For y While

```

p=1;
Do[p=p*i,{i,2006,2104,1}]
Do[If[Mod[i,17]==0,p=p*i],{i,2006,2104,1}]
p=1;
For[i=2006, i≤ 2104, i= i+17, p=p*i]
For[i=2006, i≤ 2104, i= i+17,If[Mod[i,17]==0,p=p*i]]

```

Out[*]=

73 850 604 651 472 793 760

Ejercicio 4.3

Siguiendo la notación del ejercicio 3.3

```

In[1]:= d1d2 = 03;
m1m2 = 06;
anyo = 2004;
anyoPlusMes = anyo*(m1m2 + 6);

```

a) Construir un bucle que nos de todos los múltiplos de D1D2 comprendidos entre el año de tu nacimiento y el año de tu nacimiento multiplicado por M1M2 + 6.

```

In[5]:= listaAMultiplosFor={};
listaAMultiplosDo = {};
listaAMultiplosWhile={};
For[i=anyo,i≤anyoPlusMes,i++,
  If[Mod[i,d1d2]==0,
    AppendTo[listaAMultiplosFor,i];
  ]
]

i = anyo;
While[i ≤ anyoPlusMes,
  If[Mod[i, d1d2] == 0,
    AppendTo[listaAMultiplosWhile, i]
  ];
  i++;
];
Do[
  If[Mod[i, d1d2] == 0,
    AppendTo[listaAMultiplosDo, i]
  ],
  {i, anyo, anyoPlusMes}
];
If[listaAMultiplosFor === listaAMultiplosWhile === listaAMultiplosDo,
  Print[listaAMultiplosFor]
];

```

{2004, 2007, 2010, 2013, 2016, 2019, 2022, 2025, 2028, 2031, 2034, 2037, 2040, 2043, 2046, 2049, 2052, 2055, 2058, 2061, 2064, 2067, 2070, 2073, 2076, 2079, 2082, 2085, 2088, 2091, 2094, 2097, 2100, 2103, 2106, 2109, 2112, 2115, 2118, 2121, 2124, 2127, 2130, 2133, 2136, 2139, 2142, 2145, 2148, 2151, 2154, 2157, 2160, 2163, 2166, 2169, 2172, 2175, 2178, 2181,

2184, 2187, 2190, 2193, 2196, 2199, 2202, 2205, 2208, 2211, 2214, 2217, 2220, 2223, 2226,
 2229, 2232, 2235, 2238, 2241, 2244, 2247, 2250, 2253, 2256, 2259, 2262, 2265, 2268, 2271,
 2274, 2277, 2280, 2283, 2286, 2289, 2292, 2295, 2298, 2301, 2304, 2307, 2310, 2313, 2316,
 2319, 2322, 2325, 2328, 2331, 2334, 2337, 2340, 2343, 2346, 2349, 2352, 2355, 2358, 2361,
 2364, 2367, 2370, 2373, 2376, 2379, 2382, 2385, 2388, 2391, 2394, 2397, 2400, 2403, 2406,
 2409, 2412, 2415, 2418, 2421, 2424, 2427, 2430, 2433, 2436, 2439, 2442, 2445, 2448, 2451,
 2454, 2457, 2460, 2463, 2466, 2469, 2472, 2475, 2478, 2481, 2484, 2487, 2490, 2493, 2496,
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 2634, 2637, 2640, 2643, 2646, 2649, 2652, 2655, 2658, 2661, 2664, 2667, 2670, 2673, 2676,
 2679, 2682, 2685, 2688, 2691, 2694, 2697, 2700, 2703, 2706, 2709, 2712, 2715, 2718, 2721,
 2724, 2727, 2730, 2733, 2736, 2739, 2742, 2745, 2748, 2751, 2754, 2757, 2760, 2763, 2766,
 2769, 2772, 2775, 2778, 2781, 2784, 2787, 2790, 2793, 2796, 2799, 2802, 2805, 2808, 2811,
 2814, 2817, 2820, 2823, 2826, 2829, 2832, 2835, 2838, 2841, 2844, 2847, 2850, 2853, 2856,
 2859, 2862, 2865, 2868, 2871, 2874, 2877, 2880, 2883, 2886, 2889, 2892, 2895, 2898, 2901,
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 3444, 3447, 3450, 3453, 3456, 3459, 3462, 3465, 3468, 3471, 3474, 3477, 3480, 3483, 3486,
 3489, 3492, 3495, 3498, 3501, 3504, 3507, 3510, 3513, 3516, 3519, 3522, 3525, 3528, 3531,
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 3579, 3582, 3585, 3588, 3591, 3594, 3597, 3600, 3603, 3606, 3609, 3612, 3615, 3618, 3621,
 3624, 3627, 3630, 3633, 3636, 3639, 3642, 3645, 3648, 3651, 3654, 3657, 3660, 3663, 3666,
 3669, 3672, 3675, 3678, 3681, 3684, 3687, 3690, 3693, 3696, 3699, 3702, 3705, 3708, 3711,
 3714, 3717, 3720, 3723, 3726, 3729, 3732, 3735, 3738, 3741, 3744, 3747, 3750, 3753, 3756,
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 5694, 5697, 5700, 5703, 5706, 5709, 5712, 5715, 5718, 5721, 5724, 5727, 5730, 5733, 5736,
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 6054, 6057, 6060, 6063, 6066, 6069, 6072, 6075, 6078, 6081, 6084, 6087, 6090, 6093, 6096,
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 6144, 6147, 6150, 6153, 6156, 6159, 6162, 6165, 6168, 6171, 6174, 6177, 6180, 6183, 6186,
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 6324, 6327, 6330, 6333, 6336, 6339, 6342, 6345, 6348, 6351, 6354, 6357, 6360, 6363, 6366,
 6369, 6372, 6375, 6378, 6381, 6384, 6387, 6390, 6393, 6396, 6399, 6402, 6405, 6408, 6411,
 6414, 6417, 6420, 6423, 6426, 6429, 6432, 6435, 6438, 6441, 6444, 6447, 6450, 6453, 6456,
 6459, 6462, 6465, 6468, 6471, 6474, 6477, 6480, 6483, 6486, 6489, 6492, 6495, 6498, 6501,
 6504, 6507, 6510, 6513, 6516, 6519, 6522, 6525, 6528, 6531, 6534, 6537, 6540, 6543, 6546,
 6549, 6552, 6555, 6558, 6561, 6564, 6567, 6570, 6573, 6576, 6579, 6582, 6585, 6588, 6591,
 6594, 6597, 6600, 6603, 6606, 6609, 6612, 6615, 6618, 6621, 6624, 6627, 6630, 6633, 6636,
 6639, 6642, 6645, 6648, 6651, 6654, 6657, 6660, 6663, 6666, 6669, 6672, 6675, 6678, 6681,
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23 547, 23 550, 23 553, 23 556, 23 559, 23 562, 23 565, 23 568, 23 571, 23 574, 23 577, 23 580,
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b) Usar un bucle para crear una lista con los 25 primeros múltiplos de $D1D2 + M1M2$.

```
In[13]:= diaMes=(d1d2+m1m2);
listaBMultiplosFor={};
listaBMultiplosWhile={};
listaBMultiplosDo = {};
For[i = 1, i ≤ 25, i++,
  AppendTo[listaBMultiplosFor, i * diaMes];
];
i = 1;
While[i ≤ 25,
  AppendTo[listaBMultiplosWhile, i * diaMes];
  i++;
];
Do[
  AppendTo[listaBMultiplosDo, i * diaMes],
  {i, 1, 25}
];
If[listaBMultiplosFor === listaBMultiplosWhile === listaBMultiplosDo,
  Print[listaBMultiplosFor]
];
```

{9, 18, 27, 36, 45, 54, 63, 72, 81, 90, 99, 108,
 117, 126, 135, 144, 153, 162, 171, 180, 189, 198, 207, 216, 225}

c) Calcular el producto de los múltiplos de $D1D2 + M1M2$ comprendidos entre $A1A2A3A4$ y $A1A2A3A4 + 100$

```

In[22]:= diaMes=(d1d2+m1m2);
anyoMas100=(anyo+100);
listaCMultiplosFor=1;
listaCMultiplosWhile=1;
listaCMultiplosDo = 1;
For[i = anyo, i ≤ anyoMas100, i++,
  If[Mod[i, diaMes] == 0,
    listaCMultiplosFor *= i;
  ]
];
i = anyo;
While[i ≤ anyoMas100,
  If[Mod[i, diaMes] == 0,
    listaCMultiplosWhile *= i;
  ] ×
  i++;
];
Do[
  If[Mod[i, diaMes] == 0,
    listaCMultiplosDo *= i;
  ],
  {i, anyo, anyoMas100}
];
If[listaCMultiplosFor === listaCMultiplosWhile === listaCMultiplosDo,
  Print[listaCMultiplosFor]
];

```

2 713 259 615 850 273 646 479 903 734 601 216 000

d) Calcular la suma de los múltiplos de $D1D2 + M1M2 + 10$ comprendidos entre $A1A2A3A4$ y $(A1A2A3A4)^2$.

```

In[32]:= diaMesMas10=(d1d2+m1m2+10);
anyoElev2=(anyo^2);
sumaMultiplosFor=0;
sumaMultiplosWhile=0;
sumaMultiplosDo=0;
For[i = anyo, i ≤ anyoElev2, i++,
  If[Mod[i, diaMesMas10] == 0,
    sumaMultiplosFor += i;
  ]
];
i = anyo;
While[i ≤ anyoElev2,
  If[Mod[i, diaMesMas10] == 0,
    sumaMultiplosWhile += i;
  ] ×
  i++;
];
Do[
  If[Mod[i, diaMesMas10] == 0,
    sumaMultiplosDo += i;
  ],
  {i, anyo, anyoElev2}
];
If[sumaMultiplosFor === sumaMultiplosWhile === sumaMultiplosDo,
  Print[sumaMultiplosFor]
];

```

424 432 016 800